Statistical-Mechanical and Predictive Diagnostics for Swarm Misalignment

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Research areas: Multi-agent dynamics

Research question:

Can statistical-mechanical observables provide reliable evaluations and early-warning diagnostics for emergent misalignment in multi-agent AI systems?

Why it matters:

The Hugging Face incident provided us with an extremely legible example of emergent swarm misalignment, and as such, we consider that understanding the characteristics of the swarm dynamics which enabled and exacerbated the problem is a pressing matter. Recent work has made apparent the deep links between Game-Theory and Statistical Mechanics [1], and we seek to empirically understand whether the latter framework is applicable to understanding how misalignment emerges in an agent swarm, and whether we can use it to gain insight into predictive diagnostics in swarm alignment.

The use of statistical mechanics in theoretical research for AI-Safety is not without precedent. Indeed, we consider that much theoretical work on Deep Learning Theory and phenomenology therein has been couched within the language and formalism of statistical physics (see [2] for a good specific example on grokking, and [3] for broad-based review).

We hypothesise that collective observables may be less sensitive to the evaluation-awareness of individual agents [5] than per-agent behavioural tests. We do not assume this: agents that detect monitoring could also alter population-level statistics, and we will test it directly. We will investigate whether observables such as misalignment susceptibility, temporal persistence, and inter-agent correlations can provide robust diagnostics of emergent collective behaviour, and whether these diagnostics generalise across agent populations, interaction topologies (Which other agents a given agent plays the game with), and task environments.

Background:

[1] Bouchaffera, Ykhlef, Lebbah, and Azzag, *A Collective Variational Principle Unifying Bayesian Inference, Game Theory, and Thermodynamics*. This paper provides us with our starting link from swarm dynamics to statistical mechanics.

[2] Rubin, Seroussi and Ringel, *Grokking as a First Order Phase Transition in Two Layer Networks*. This work studies feature learning in teacher–student models using an adaptive-kernel framework and relates grokking to first-order phase-transition phenomenology. Its modular-arithmetic setting provides an analytically tractable starting point for investigating the relationship between learned features, correlators and scaling behaviour.

[3] Simon, Kunin, et al. *There Will Be a Scientific Theory of Deep Learning*. This paper provides us a review of relevant theoretical advances in the theory of deep learning with a particular regard for the applicability of statistical mechanics.

[4] Henkel, Hinrichsen, Lubeck, *Non-Equilibrium Phase Transitions.* This textbook provides us with a standard reference for the canonical techniques and reference values used in understanding and approximating non-equilibrium phase transitions.

[5] Goldowsky-Dill, Balesni, Scheurer, Hobbhahn, *Claude Sonnet 3.7 (often) knows when it's in alignment evaluations*. This blog-post provides evidence that sufficiently advanced AIs are aware of when they are being evaluated.

Approach and Setup:

We begin with our working hypothesis being the claim that swarm misalignment may exhibit the scaling properties (e.g. power law relations) of a mean-field Ising model in the equilibrium case, and directed percolation in the dynamical case. We will interrogate this claim by constructing a host of multi-agent systems, and examining how properties of the agent population change as misalignment emerges.

We now detail how we expect the Ising-Misalignment correspondence to be drawn. We consider N symmetric two-action agents, with actions mapped to aligned ($s = +1$) and misaligned ($s = -1$). Payoffs follow a symmetric 2×2 game (T, R, P, S) plus an alignment incentive a added to the aligned action's payoff. For selfish agents the game is a potential game, with effective field $h = (R+S-T-P)/4 + a/2$ and pair coupling $J = (R-S-T+P)/4$. For a standard Prisoner's Dilemma, $h < 0$ at $a = 0$, so there is no critical point at zero incentive. We use this as a negative control. Criticality requires $J > 0$ ($R+P > S+T$) and tuning a to $a^* = (T+P-R-S)/2$ so that $h = 0$. The incentive a plays the role of the conjugate field, and the update noise β plays the role of inverse temperature. These are the two control parameters we vary.

Hypothesis 1 (equilibrium). Under homogeneous (all at once) updating on a complete graph (where the mean-field assumption will hold), with J scaled by 1/N, the order parameter m, susceptibility $\chi = \partial m/\partial a$, and persistence time follow mean-field Ising scaling near the tuned critical point.

Hypothesis 2 (dynamical). Under imitation updating with no spontaneous re-alignment (MDPs update their policies based on the behaviour of other agents), the dynamics have an absorbing state, and the transition is in the directed-percolation class. We run two variants (again initially on complete connected graphs): **(a)** all-misaligned absorbing (alignment extinction), and **(b)** all-aligned absorbing (misalignment as contagion), which is closer to the Hugging Face incident's peer-influence structure. We run both and report which fits. We also note here that we have taken a necessarily simple coarse-graining of actual misalignment, whereby we assume any agent collaborating with, or tacitly accommodating a misaligned agent itself to have chosen to be in the misaligned state.

For each hypothesis we state in advance the reference exponents [4]. Each is refuted if the order parameter, susceptibility, and persistence statistics fail to collapse onto the reference scaling functions across N.

Beyond the above-described setup and tasks, we hope to extend our experimental setup to examine similar such phenomena in more capable models (likely necessitating an extension of the theoretical framework). In this case, the task will extend beyond the well defined game-theoretic structure of prisoners' dilemma, to less constrained environments in which agents must infer their objectives, interact with other agents, and decide whether to pursue a shared objective or a conflicting one. We note that this is difficult to design, and that this extension is largely speculative. We will investigate whether the scaling relations observed in our minimal studies persist as we vary model capability, population size, interaction topology, and incentives for alignment. In particular, we will measure the mean misalignment fraction, its susceptibility to changes in the incentive structure, the temporal persistence of misaligned states, and the emergence of correlated or coordinated misalignment across the population.

What success looks like:

We organise our project into three milestones, each with distinct evaluation-oriented deliverables.

1. Measurement framework and tests of Hypothesis 1&2**.** Build a reproducible simulator and operational definitions of alignment and misalignment. (a) Verify the estimators for m, χ, correlation functions and persistence on exactly solvable Ising and DP simulations. (b) Test whether the swarm's observables collapse onto either class (H1, H2) under the stated criteria, including the PD negative

control. *Deliverables:* simulator, estimator validation suite, and a report of confirmed or refuted scaling hypotheses.

2. Predictive value of collective diagnostics. Define a misalignment event as a minimal time duration of at least τ steps where m crosses a threshold m*. We vary N, incentive strength, noise, and topology. Compare the incremental predictive value of susceptibility, correlation, and persistence observables against three baselines: aggregate misalignment-fraction trends, generic early-warning indicators (rising variance and autocorrelation), and a change-point or hidden-state filtering detector. Report sensitivity, specificity, false-positive rate, and lead time. We shall hold out topology and N as separate generalisation tests.
3. Test generalisation to capable models. Apply the validated evaluation framework to a small, carefully controlled population of language-model agents. Define misalignment relative to explicit task objectives and independently specified behavioural constraints. Investigate whether the diagnostics developed in the scripted-agent regime remain informative across model families, capability levels, and interaction structures. The deliverable will be an initial transfer evaluation assessing the generalisation of our proposed diagnostics beyond analytically tractable agents. We note that by virtue of the resource intensive nature of such an exercise, much of our discussion around this remains necessarily speculative for reasons outlined in the following section.

Over the course of a three-month fellowship, we expect to complete the first two milestones. The third represents a substantially more challenging extension, both in terms of both computational requirements and safety protocols. As such, we leave that as a stretch goal for follow on research. We consider that the main difficulty involves the establishment of adequate safety protocols necessary for evaluating capable agents, and we detail these difficulties in the proceeding section.

Risks, Challenges, and open questions:

Risks:

1. Imperfect knowledge on the part of the observer in the experimental setup is highly task, and setup dependent, and we can identify three main failure modes in this avenue via construct, measurement, and predictive validity. The first is whether or not the candidate universality classes we are examining are reasonably good models of actual misalignment. Secondly, there is an open question around how reliably an evaluator can infer an agent's alignment state from observable behaviour. Finally, we ask whether the resulting measurements predict future collective misalignment, rather than merely correlate with behaviour already observed?

2. We consider now one very real risk of the generalization task to more capable models. Indeed, if the alignment task evaluation is not sufficiently well specified, the experimental setup not sufficiently sandboxed or the agents not made sufficiently benign, then there is a real possibility of a second hugging-face-type incident occurring while trying to collect sufficient data on realistic agents. We consider that this will require models to run in isolated environments with no network access.
3. If misalignment onset is abrupt (first-order or hybrid), scaling-based warnings may be weak or absent. Our grokking reference [2] is itself a first-order result. We therefore do not assume second-order behaviour: Milestone 1 includes tests that discriminate the order of the transition (hysteresis, bimodal order-parameter distributions, Binder cumulants), and a first-order finding would be reported as a result, with the diagnostics re-scoped accordingly.
4. Universality requires large N and scaling collapse; capable-agent populations will be small in no small part due to budgetary and computational constraints. Milestone 3 results are therefore limited to qualitative signatures, not exponent estimates.

Open Questions:

The Ising model provides us with a tractable, if over-simplified approximation of emergent swarm misalignment whereby an agent may either be totally misaligned, or totally aligned. In reality an agent may possess a variety of discrete personas, each with varying degrees of misalignment and perhaps in a variety of different non-uniform ways, indeed, the numbers of personas likely varies per agent, and per model. One may consider a non-uniform generalization of the Potts model, or an O(N) model, and dynamical percolation type transitions in a more general setting than just the model we would be considering.

The Two-state system setup also implies a perfect knowledge from an observer of whether a system of agents, or indeed a single agent, is or isn't aligned. To this end we consider that beyond more complex higher dimensional cases for misalignment, then there needs to be a systematic way of incorporating this into our theoretical results to make them match experiments more closely, in a framework which retains the integrity of the original results. A possible solution might involve examining and probing the belief geometry of such agents directly.

If our research proves successful, we might naturally ask whether we can use the established links between statistical mechanics and game theory to design optimal swarm interventions, such as unstable games which structurally prevent swarm misalignment from emerging at all? Examples of this include anti-aligned couplings (for

example in the antiferromagnetic Ising model on a triangular lattice where no ordered state emerges), more restrictive swarm topology, or injection of noise into agent communications, degrading the reliability of their actions. We might hope for these to have similarly predictive results.